

NAMAN SAMADHIYA

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SUMMARY

Pre-final year Electronics Engineering (VLSI Design & Technology) student at VIT Vellore with experience in semiconductor device research, RTL design, FPGA development, and hardware verification. Skilled in Verilog/SystemVerilog and EDA tools, with hands-on experience in TCAD-based device analysis and digital hardware development.

EDUCATION

Vellore Institute of Technology

B.Tech in Electronics Engineering (VLSI Design and Technology) – CGPA: 7.77

Vellore, India

2024 – Present

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Python, Tcl, Perl, Java

Digital Design: RTL Design, Digital Logic, FSM Design, FPGA Development, Hardware Verification

EDA Tools: Cadence Virtuoso, Xcelium, Tempus, ModelSim, Quartus Prime, Silvaco ATLAS, LTspice

Tools & Platforms: Git, GitHub, ESP32, Arduino, UART

EXPERIENCE

Madhav Institute of Technology & Science (MITS)

May – Aug 2026

VLSI Research Intern

- Conducted semiconductor device research on Vertical TFETs for ultra-low-power biosensing applications using Silvaco ATLAS TCAD.
- Designed and analyzed an InGaAs/GaAs heterostructured Vertical TFET biosensor, evaluating I_{ON} , I_{OFF} , threshold voltage, and subthreshold swing under varying dielectric constants.

PROJECTS

Natural Language–Driven SystemVerilog Generation & Verification Framework

Python, SystemVerilog, Flask, ModelSim

- Developed an AI-assisted framework to generate synthesizable SystemVerilog RTL and self-checking testbenches from natural language specifications.
- Automated RTL generation, reference modeling, and ModelSim-based verification into an integrated hardware design workflow.

Smart Insole – Real-Time Foot Pressure Monitoring & Analysis

Piezoelectric Sensors, Energy Harvesting, Bluetooth, ML

- Developed a piezoelectric smart insole for real-time foot-pressure monitoring across five regions, with integrated energy harvesting from footstep-induced mechanical stress.
- Implemented Bluetooth-based data transmission, pressure visualization, historical tracking, and ML-based analysis for detecting conditions such as flatfoot and gait imbalance.

Stop-and-Go Pedestrian Crossing Signal Controller

Verilog HDL, FPGA, ModelSim

- Implemented an FPGA-based pedestrian crossing controller using a Finite State Machine (FSM) architecture.
- Verified RTL functionality through ModelSim simulation and validated the design through real-time FPGA implementation.

LEADERSHIP & ACTIVITIES

The Cubing Club, VIT Vellore

Jan 2026 – Present

Vice Chairperson

- Lead planning and execution of technical events, competitions, and club initiatives through cross-functional team coordination.
- Manage recruitment, team operations, and event logistics to support effective execution of club activities.

V-SPACE, VIT Vellore

Jul 2026 – Present

Core Member – Events

- Plan, coordinate, and execute engaging events while collaborating with team members to enhance participant experiences.

IEEE Circuits and Systems Society (CASS) – Madras Chapter

Aug 2026 – Present

Official Student Volunteer

- Contribute to technical activities and event coordination as an official volunteer of the IEEE CASS Madras Chapter.

DVCon India 2026

Sep 2026

Volunteer

- Volunteered at DVCon India 2026, supporting conference coordination and event operations.

SmowXpress IoT & Smowcode Workshop, VIT Vellore

Sep 2025

Point of Contact

- Coordinated an ESP32-based IoT workshop for 120+ participants and assisted with live project sessions.

CERTIFICATIONS

• **Maven Silicon** – VLSI System On Chip Design – Overview (SoC architecture & VLSI flow).

Certificate

• **VLSI FOR ALL** – VLSI SoC Design using Verilog HDL Workshop (RTL modeling, FSM design).

Certificate

• **Qualcomm Academy** – AI Upskilling Certificate: Technical Foundations (Generative AI, Edge AI, Qualcomm AI Hub).

Certificate

ACHIEVEMENTS

- Secured 4th place among 60+ teams at Makeathon 2026 for developing a cooperative autonomous dual-vehicle V2V system.
- Co-developed a smart orthopedic pressure-analysis insole system patent filing in progress